

Compiler Design

Lexical Analyzer

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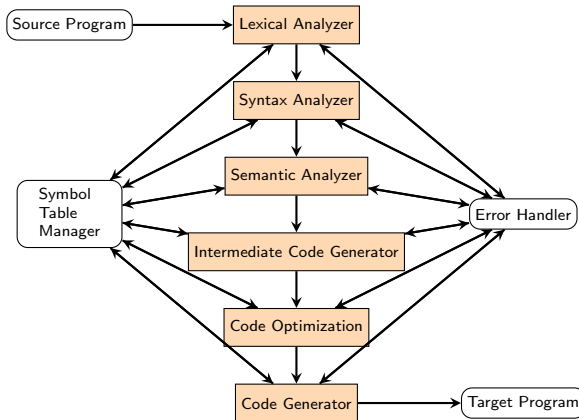
3 Specification of Tokens

- Regular Expressions
- Deterministic Finite Automata (DFA)
- Nondeterministic Finite Acceptor (NFA)
- Nondeterministic Finite Acceptor with ϵ -transitions (ϵ -NFA)

4 Recognitions of Tokens

The Phases of a Compiler

- Conceptually, a compiler operates in phases, each of which translates the source program from one representation to another.



Lexical Analyzer

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- Stripping from the source program comments and white space in the form of blank, tab, and newline characters.
- Correlating error messages from the compiler with the same source program

Lexical Analyzer

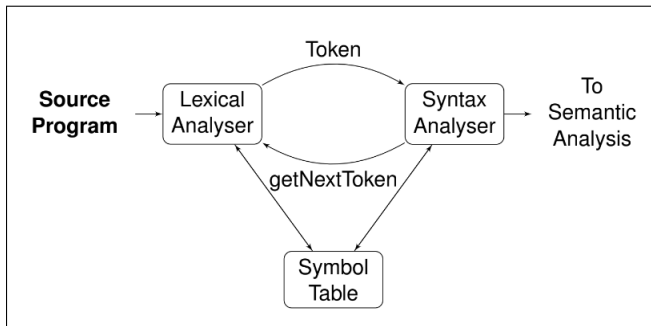


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Tokens, Patterns and Lexemes

- A **token** is a pair consisting of a token name and an optional attribute value. The token name is an abstract symbol representing a kind of lexical unit, e.g., a particular keyword, or a sequence of input characters denoting an identifier.

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- This set of strings is described by a rule called **pattern** associated with that token. The pattern is said to match each string in the set.

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- This set of strings is described by a rule called **pattern** associated with that token. The pattern is said to match each string in the set.
- A **lexeme** is a sequence of characters in the source program that matches the pattern for a token and is identified by the lexical analyser as an instance of that token. These are smallest logical unit (words) of a program such as A, B, 1.0, true, +, j=

Examples - Tokens, Patterns and Lexemes

Consider The Following C Statement:

```
printf ("Total = %d", score) ;
```

- **printf** and **score** are lexemes that match the pattern for the token **id**
- "Total = %d" is a lexeme matching literal.

Token	Sample lexemes	Pattern
if	if	Characters i, f
else	else	Characters e, l, s, e
comparison		
id	pi, score, d2	letters followed by letters and digit
number		any numeric constant
literal	"Total = %d"	Total = %d

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Specification of Tokens

- Regular Expression.
- Deterministic Finite Automata.
- Non-Deterministic Finite Automata.
- Non-Deterministic Finite Automata with empty transitions.

Regular Expressions

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- **Regular Expression** is a way to represent a language.
- **Regular expressions** describe **Regular language**.
- **Example:** $(a + bc)^*$
describes the language
 $\{a, bc\}^* = \{\lambda, a, bc, aa, abc, bca, \dots\}$

Regular Expression

- **Recursive Definition:**

- Primitive regular expressions: Φ, λ, α

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- A regular expression: $(a + bc)^*(c + \Phi)$
- Not a regular expression: $(a+b+)$

Languages of Regular Expressions

- $L(R)$: language of regular expression, r

Example: $L((a + b + c)^*) = \{\lambda, a, bc, aa, abc, bca, \dots\}$

Languages of Regular Expressions

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Example: $L((a + b + c)^*) = \{\lambda, a, bc, aa, abc, bca, \dots\}$

- For primitive regular expressions:

- $L(\Phi) = \Phi$
- $L(\lambda) = \{\lambda\}$
- $L(a) = \{a\}$
- $L(r_1 + r_2) = L(r_1) \cup L(r_2)$
- $L(r_1.r_2) = L(r_1)L(r_2)$
- $L(r_1^*) = (L(r_1))^*$
- $(L(r_1)) = L(r_1)$

Languages of Regular Expressions

- **Example:**

Regular Expression: $(a + b).a^*$

$$\begin{aligned} L((a + b).a^*) &= L(a + b)L(a^*) = (L(a) \cup L(b))L(a^*) = \\ &(\{a\} \cup \{b\})\{(a)^*\} = \{a, b\}\{\lambda, a, aa, aaa, \dots\} = \\ &\{a, aa, aaa, \dots, b, bb, bbb, \dots\} \end{aligned}$$

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- **Regular Expression:** $r = (0 + 1)^*00(0 + 1)^*$

$$L(r) = \{\text{all strings with at least two consecutive } 0\}$$

- **Regular Expression:** $r = (1 + 01)^*(0 + \lambda)$

$$L(r) = \{\text{all strings without two consecutive } 0\}$$

Equivalent Regular Expressions

- Regular expressions r_1 and r_2 are equivalent if $L(r_1) = L(r_2)$
- **Example:** $L = \{ \text{all strings without two consecutive 0} \}$
 $r_1 = (0 + 01)^*(0 + \lambda)$
 $r_2 = (1^*011^*)^*(0 + \lambda) + 1^*(0 + \lambda)$
 $L(r_1) = L(r_2) = L \implies r_1 \text{ and } r_2 \text{ are equivalent regular expression.}$

Regular Expressions and Regular Languages

Languages Generated by
by
Regular Expressions } = { **Regular Languages** }

Regular Expressions and Regular Languages

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Regular Expressions $\Bigg\} = \{\textbf{Regular Languages}\}$

- **Theorem 1:** For any regular expression r the language $L(r)$ is regular.
- **Theorem 2:** For any regular language, $L(r)$, there is a regular expression r with $L(r) = L$.

Deterministic Finite Automata (DFA)

- **Deterministic Finite Automata (DFA):**

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q : Finite set of states

Σ : Input alphabet

δ : Transition Function: $\delta : Q \times \Sigma \rightarrow Q$

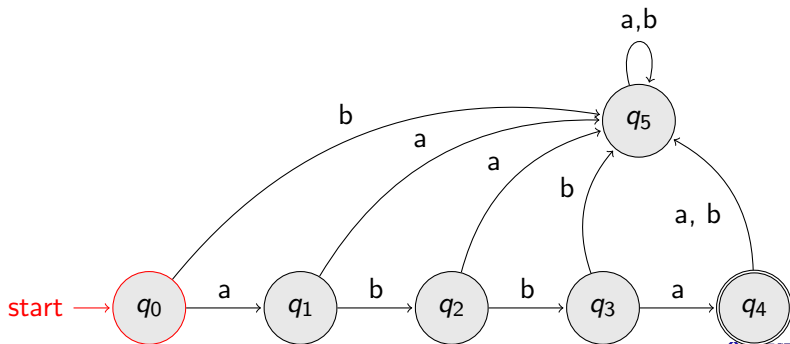
q_0 : Initial State

F : Finite set of final states: $F \subseteq Q$

Formalities

- **Transition Function** , δ :

$$\delta : Q \times \Sigma \rightarrow Q$$

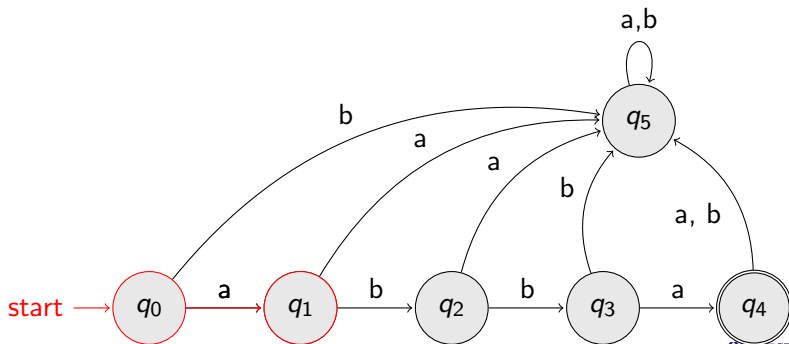


Formalities

- **Transition Function** , δ :

$$\delta : Q \times \Sigma \rightarrow Q$$

$$\delta(q_0, a) = q_1$$



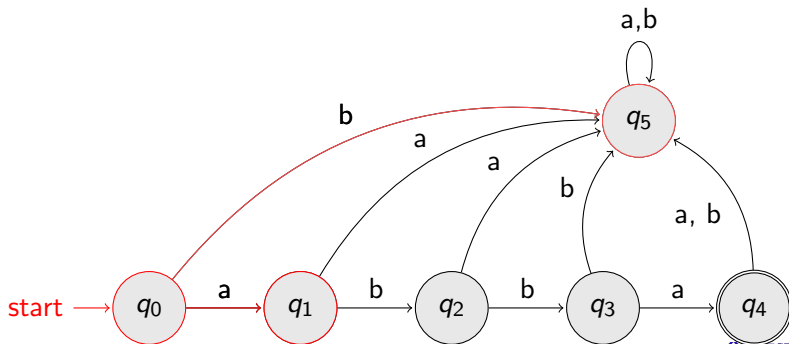
Formalities

- **Transition Function** , δ :

$$\delta : Q \times \Sigma \rightarrow Q$$

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_5$$



Formalities

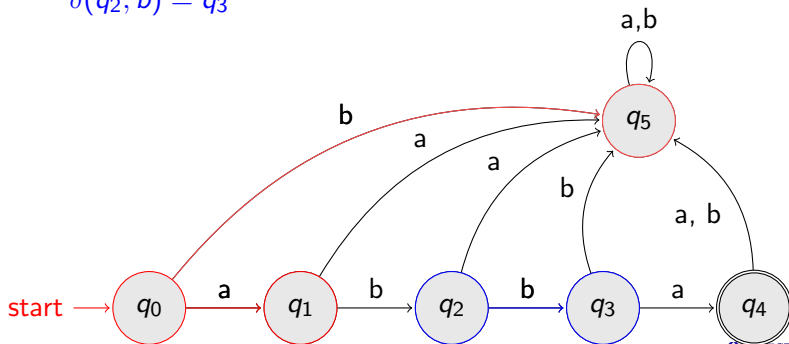
- Transition Function , δ :

$$\delta : Q \times \Sigma \rightarrow Q$$

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_5$$

$$\delta(q_2, b) = q_3$$

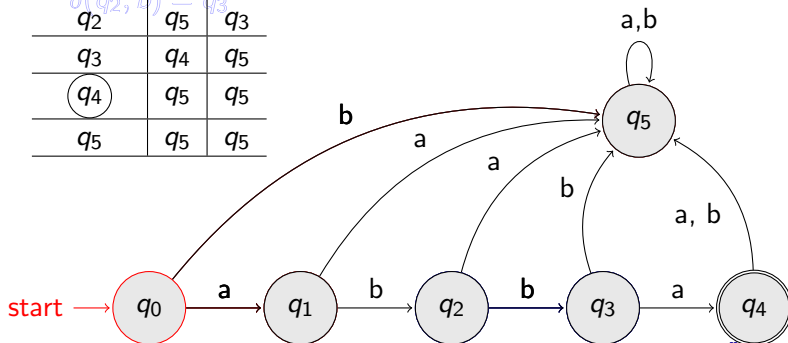


Formalities

- Transition Function , δ :

$$\delta : Q \times \Sigma \rightarrow Q$$

States	a	b
q_0	q_1	q_5
q_1	q_5	q_2
q_2	q_5	q_3
q_3	q_4	q_5
q_4	q_5	q_5
q_5	q_5	q_5



Languages Accepted by DFAs

- The language accepted by a DFA, $M = (Q, \Sigma, \delta, q_0, F)$ can be defined as follows:

- $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$

Here,

Q : Finite set of states

Σ : Input alphabet

δ : Transition function: $\delta^* : Q \times \Sigma^* \rightarrow Q$

q_0 : Initial State

F : Finite set of final states

Nondeterministic Finite Acceptor (NFA)

- **Nondeterministic Finite Acceptor (NFA):**

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q : Finite set of states

Σ : Input alphabet

δ : Transition Function: $\delta : Q \times \Sigma \rightarrow 2^Q$

q_0 : Initial State

F : Finite set of final/accepting states: $F \subseteq Q$

Nondeterministic Finite Acceptor with ϵ -transitions (ϵ -NFA)

- **Nondeterministic Finite Acceptor with ϵ -transitions (ϵ -NFA):**

$$M = (Q, \Sigma \cup \{\epsilon\}, \delta, q_0, F)$$

Q : Finite set of states

$\Sigma \cup \{\epsilon\}$: Input alphabet, $\{\epsilon\}$ is a member of this alphabet.

δ : Transition Function: $\delta : Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$

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Recognitions of Tokens

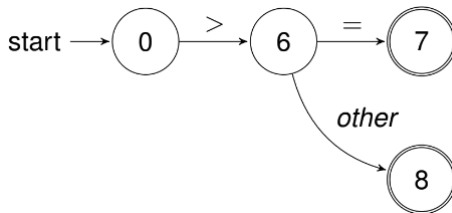
Regular expression pattern

Regular Expression	Token	Attribute Value
WS	–	–
if	if	–
then	then	–
else	else	–
id	id	pointer to table entry
num	num	pointer to table entry
<	relop	LT
<=	relop	LE
>	relop	GT
<=	relop	LE
=	relop	EQ
<>	relop	NE

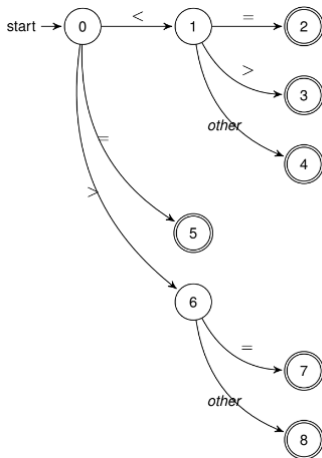


Construct a lexical analyser that will isolate the lexeme for the next token in the input buffer and produce as output a pair consisting of the appropriate token and attribute-value, using the given translation table.

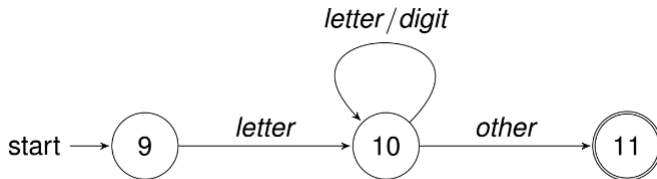
Transition Diagram for $\geq$



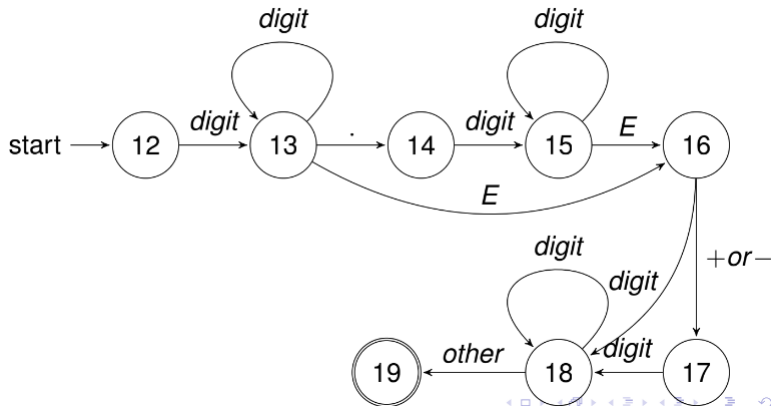
Transition Diagrams for Relational Operators



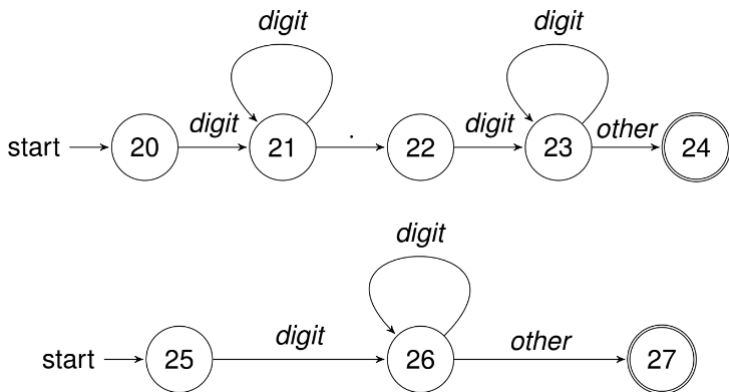
Transition Diagrams for Identifiers or Keywords



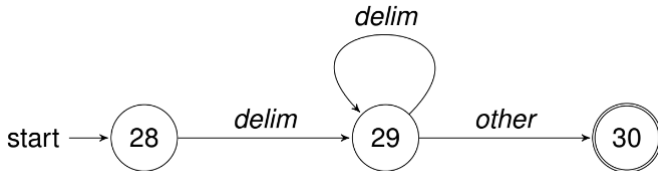
Transition Diagram for Numbers



Transition Diagram for Numbers



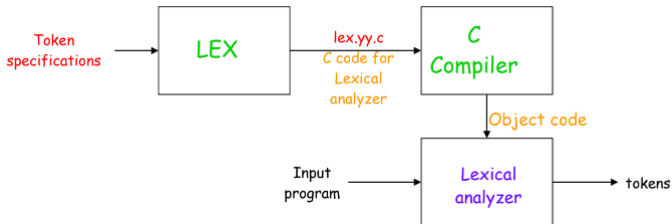
Transition Diagrams for White spaces



Implementing a Transition Diagram

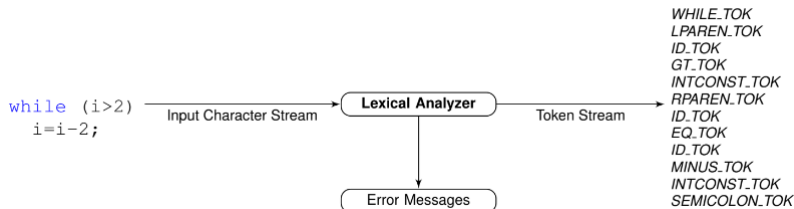
```
token nexttoken()
{ while (1) {
    switch (state) {
    case 0: c = nextchar();
        if (c==blank || c==tab || c==newline) {
            state = 0;
            lexeme_beginning++;
        }
        else if (c=='<') state = 1;
        else if (c=='=') state = 5;
        else if (c=='>') state = 6;
        else state = fail();
        break;
    case 1:
        ...
    case 9: c = nextchar();
        if (isletter(c)) state = 10;
        else state = fail();
        break;
    case 10: c = nextchar();
        if (isletter(c)) state = 10;
        else if (isdigit(c)) state = 10;
        else state = 11;
        break;
    ...
}
```

Lexical Analyzer Generators - Lex



Lexical Analyzer Generators - Lex

- converts the input program into a sequence of Tokens.
- can be implemented with the help of Finite Automata.



Lexical Analyzer Generators - Lex

```
FILE *yyin;
char *yytext;
main(int argc, char *argv[]){
    int token;
    if (argc != 2){
    }else{
        yyin = fopen(argv[1], "r");
        while(!feof(yyin)){
            token = yylex();
            printf("%d", token);
        }
        fclose(yyin);
    }
}
```

```
int yylex(){
    ...
    ...
}
```



Lexical Analyzer Generators - Lex

Loop and switch Approach

```
/* Single caharacter lexemes */
#define LPAREN_TOK '('
#define GT_TOK '>'
#define RPAREN_TOK ')'
#define EQ_TOK '='
#define MINUS_TOK '-'
#define SEMICOLON_TOK ';'
/*.....
.....*/
/* Reserved words */
#define WHILE_TOK 256
/*.....
.....*/
/* Identifier, constants..*/
#define ID_TOK 350
#define INTCONST 351
/*.....
.....*/
```



- Alfred V. Aho, Ravi Sethi, Jeffrey D Ullman, “Compilers Principles Techniques and Tools”, Pearson Education.